

CMAN-5

Given $C = A \times B$ 2×2 Matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

expected $C = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$

TASKS:-

1.

col 0

col 1

Row 0 $PE[0][0]$

→ weight = 5

$B[0][0]$

↓ Psum

Row 1 $PE[1][0]$

→ weight = 7

$B[1][0]$

↓

output

$PE[0][1]$

→ weight = 6

$B[0][1]$

↓ Psum

$PE[1][1]$

→ weight = 8

$B[1][1]$

↓

output

Inputs from
A stream in
from left,
and partial
sums accumulate
downward

Weights from matrix B are pre-loaded into the PEs and remain fixed.

2.

Cycle	Row 0	Row 1	PE[0][0]	PE[0][1]	PE[1][0]	PE[1][1]	Output C
1	A[0][0] = 1 1	-	1x5 = 5	1x6 = 6	0	0	-
2	A[1][0] = 3 3	A[0][1] = 2 2	3x5 = 15	3x6 = 18	5+2x7 = 19	6+2x8 = 22	C[0][0] = 19 C[0][1] = 22
3	-	A[1][1] = 4 4	0	0	15+4x7 = 43	18+4x8 = 50	C[1][0] = 43 C[1][1] = 50
4	1						

$$C[0][0] = 19$$

$$C[0][1] = 22$$

$$C[1][0] = 43$$

$$C[1][1] = 50$$

$$C = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix} \text{ final output.}$$

3. a. MAC Count total

$$4 \text{ PEs} \times 2 \text{ inputs each} \\ = 8 \text{ MACs}$$

b. Input reuse

For a $N \times N$ weight an input is fetched once and reused $N-1$ times

For a 2×2 array an input is reused $2-1=1$ time.

So, they are fetched once used twice.

c7. The fetched data of the inputs gets streamed while the inputs weights stay stationary.

Matrix	Access Type	Accesses
A	Read	4
B	Read	4
C	Write	4

Total 12 off chip accesses.

&

4. Output stationary here, the partial sums (outputs) stay fixed in the PEs for the entire computation.